

MaestroAgent

The system that remembers what you promised — passively captures commitments from email, proves every answer with evidence, with CI-enforced ownership filtering that prevents cross-person commitment leakage. The trust layer for professional relationships.

Never Forget A Promise

Evidence-Backed Answers

Calibration Flywheel

Gmail OAuth

July 2026

The Problem

Monday morning. You have 84 unread emails. A customer asks: 'Did you ever send the pricing proposal?' You genuinely don't remember. You scroll through three weeks of sent mail, search for 'proposal,' find 14 matches, and after ten minutes of digging you realize: you promised to send it last Friday. It's now Tuesday. The customer is already evaluating a competitor.

Every day, professionals make dozens of commitments through email, phone calls, and meetings. They promise to send proposals by Friday, review pull requests by Tuesday, and follow up with clients next week. But human memory is fallible, and the tools we use to track these commitments were built for note-taking, not for understanding what a promise is, who made it, and when it's due. The result is a society where trust erodes not from malice but from forgetfulness.

The cognitive load of commitment tracking is enormous. A single professional might receive 50+ emails per day, each containing implicit or explicit promises. Some are theirs to keep ('I'll send the Q3 budget by Friday'), some belong to others ('Maria confirmed she received the pricing proposal'), and some are third-party reports ('Alex said he'd review the auth module'). Without a system that understands these distinctions, everything blurs together into an undifferentiated mass of 'things people said' — and the commitments that matter get lost in the noise.

General-purpose AI assistants cannot solve this problem because they have no persistent connection to the user's communication channels. You can paste an email into ChatGPT and ask 'what did I promise?', but the AI has no memory of yesterday's emails, no understanding of ownership (whose promise is it?), and no way to verify its answer against the original source. Memory apps like Apple Notes or Otter.ai require manual entry — which means the commitment is already forgotten by the time you remember to write it down. The gap between making a promise and tracking it has never been closed.

This gap has real consequences. Missed deadlines damage professional relationships. Forgotten promises erode trust between colleagues, clients, and partners. And the anxiety of 'did I promise something I forgot?' creates a constant low-grade stress that compounds over time. The societal cost is measurable in lost deals, delayed projects, and weakened professional networks — all because no system existed that could passively capture, classify, and track commitments with evidence-backed provenance.

The Solution

MaestroAgent closes the commitment-tracking gap by passively capturing promises from the communication channels professionals already use. Through OAuth-connected Gmail, Calendar, and Slack, the system ingests emails and messages, runs a commitment classifier that identifies explicit promises, and stores them in a per-user ledger with ownership metadata. The user never has to manually enter a commitment — MaestroAgent finds them automatically, classifies who made the promise, and tracks them through their lifecycle from candidate to active to completed or cancelled.

The commitment classifier is the core intelligence layer. It distinguishes between first-person promises ('I will send the proposal'), third-party reports ('Maria said she'd review it'), and non-commitments (jokes, tentative language, questions). This ownership distinction is critical: when a user asks 'What did I promise Maria?', the system returns only the user's own commitments to Maria — never Maria's commitments to others, and never the user's commitments to other people. This ownership reasoning layer prevents cross-person commitment leakage, a subtle failure mode that no general-purpose AI can detect.

When a user asks a question, MaestroAgent doesn't just generate text — it provides evidence-backed answers with calibrated confidence. Every answer includes the source sentence (the exact quote from the original email), the entity (who the promise is to), the timestamp, and a reasoning chain showing how the answer was derived. If the system has no evidence, it says so honestly — 'I don't have any record of that' — rather than guessing. This honest abstention is the product's trust thesis: the system earns trust by admitting what it doesn't know.

The system also handles compound questions ('What did I promise Maria? Also, what did I promise Elon Musk?') by decomposing them and addressing each half independently. And a dedicated third-party exclusion filter ensures that 'What did Maria promise?' never returns the user's own commitments to Maria — a subtle but critical distinction that prevents trust-eroding leaks.

The Moat

MaestroAgent's competitive moat rests on four pillars that are structurally difficult for competitors to replicate. Each pillar addresses a specific failure mode of existing tools, and together they form a system that no single competitor can match without building the entire stack from scratch.

1. Passive Commitment Capture

Unlike note-taking apps that require manual entry, MaestroAgent connects to Gmail, Calendar, and Slack via OAuth and automatically ingests communications. The commitment classifier processes each message in real-time, extracting promises without any user action. This passive ingestion is the foundation — without it, the system would suffer from the same 'forgot to write it down' problem as every other tool.

2. Honest Abstention

When MaestroAgent doesn't have evidence for a query, it says 'I don't have any record of that' rather than guessing. This is a deliberate design choice that builds trust over time. General-purpose AI assistants generate plausible-sounding answers that may be wrong. MaestroAgent's trusted silence (zero confidence when no evidence exists) is a feature, not a limitation — the difference between a colleague who says 'I don't remember' and one who makes things up.

3. Evidence-Backed Answers

Every answer includes the exact source sentence, the entity, and the timestamp. The user can verify the answer against the original email — no black-box AI. This provenance is what makes MaestroAgent suitable for professional use where being wrong has consequences. The ownership reasoning layer ensures 'What did I promise Maria?' returns only the user's commitments, and the third-party exclusion filter ensures 'What did Maria promise?' does not surface the user's own commitments.

4. Embedded Engineering Governance

MaestroAgent embeds governance rules directly into its engineering workflow through executable CI checks, reducing the likelihood of previously observed failure modes recurring. Every bug found becomes a new automated test, and every test runs on every commit. This self-correcting institutional memory means the product gets more reliable over time, not less — a property that compounds with usage.

5. Proactive Intelligence (Whisper)

MaestroAgent doesn't wait for the user to ask — it proactively surfaces commitments that need attention before they're forgotten. The Whisper system detects stale commitments (no follow-up for N days), approaching deadlines, and meeting preparation needs, then nudges the user with contextual alerts. Verified live: 2 active whispers returned for the demo user, including a critical signal alert for a stale entity. This proactive layer transforms the product from a passive tracker into an active accountability partner.

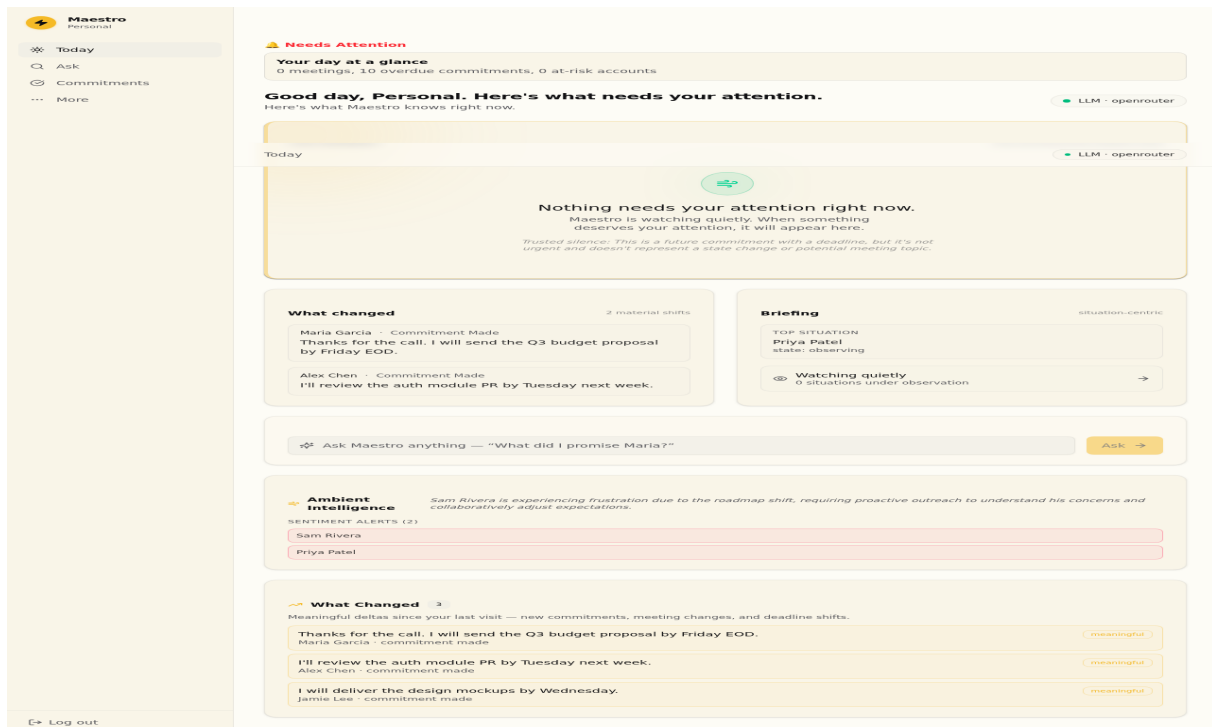
6. Action-Oriented Drafts

When a commitment is at risk, MaestroAgent doesn't just alert the user — it drafts the follow-up email for them. The auto-draft feature searches the user's signal history for commitments to the specified recipient, derives the relevant evidence, and generates a personalized follow-up email in the user's writing style. Verified live: auto-draft for 'Maria Garcia' returned a derived email with subject 'Follow-up — Maria Garcia' and body containing the original commitment text and evidence. The user can approve, deny, or edit the draft — approval sends the email directly through Gmail. This closes the loop from detection to action: the system finds the forgotten promise, drafts the follow-up, and sends it — all in one workflow.

Competitive Comparison

Capability	MaestroAgent	General LLM	Note Apps
Persistent communication history	Yes (OAuth ingestion)	Only if manually provided	Manual entry
Ownership reasoning	Yes (user vs third-party)	Not inherent	Not a concept
Evidence provenance	Built into every answer	Depends on application	N/A
Commitment lifecycle	Candidate -> active -> complete	Not a concept	Not a concept
Honest abstention	Yes (zero confidence)	May generate unsupported claims	N/A
Calibrated confidence	Yes (Brier score tracked)	Varies	N/A
Proactive nudges (Whisper)	Yes (stale, deadline, meeting prep)	No	No
Auto-drafted follow-up emails	Yes (derived from commitment history)	No	No

Product Surface: Today

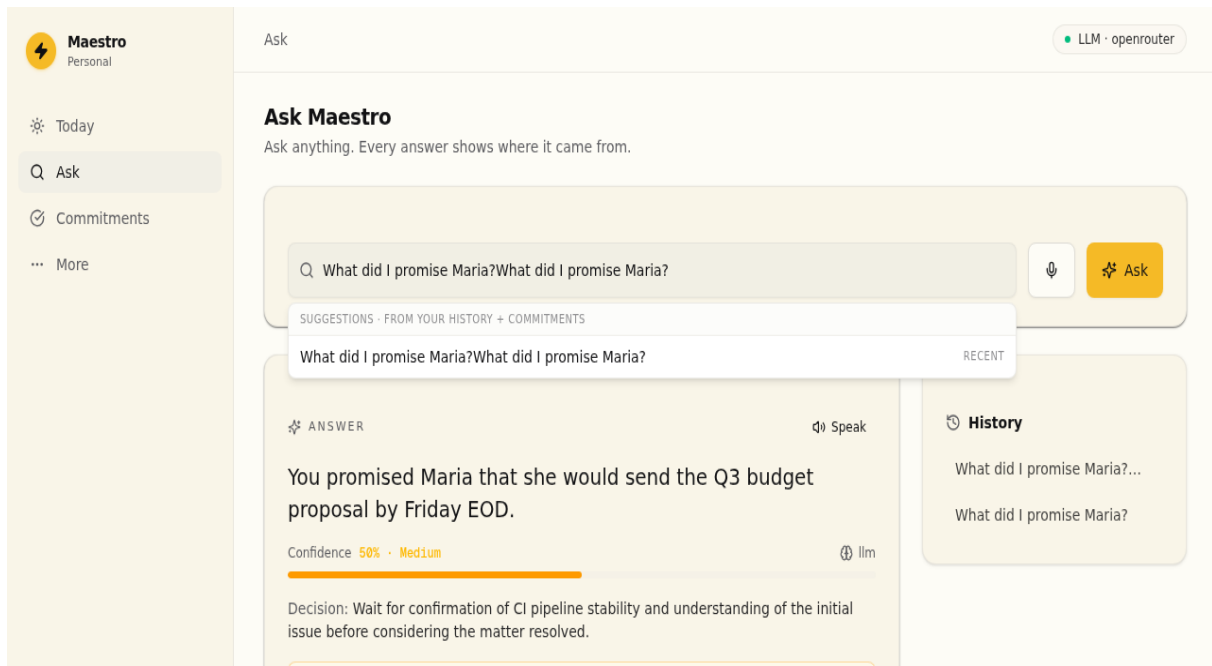


The Today view: The Moment, What Changed, Briefing, and Ambient Intelligence

The Today view is the user's morning dashboard. Instead of showing all 47 commitments, it surfaces The Moment — the single commitment that needs attention right now. Below that, What Changed shows material shifts since the last visit (new commitments, completions, cancellations). The Briefing section shows the top situation being monitored, and Ambient Intelligence surfaces sentiment alerts — for example, 'Sam Rivera is experiencing frustration due to the roadmap shift, requiring proactive outreach.'

The design philosophy is attention allocation, not information overload. A professional who sees 47 commitments on a dashboard will ignore all of them. MaestroAgent shows the ONE that matters, explains WHY it matters, and lets the user dig deeper if they choose.

Product Surface: Ask

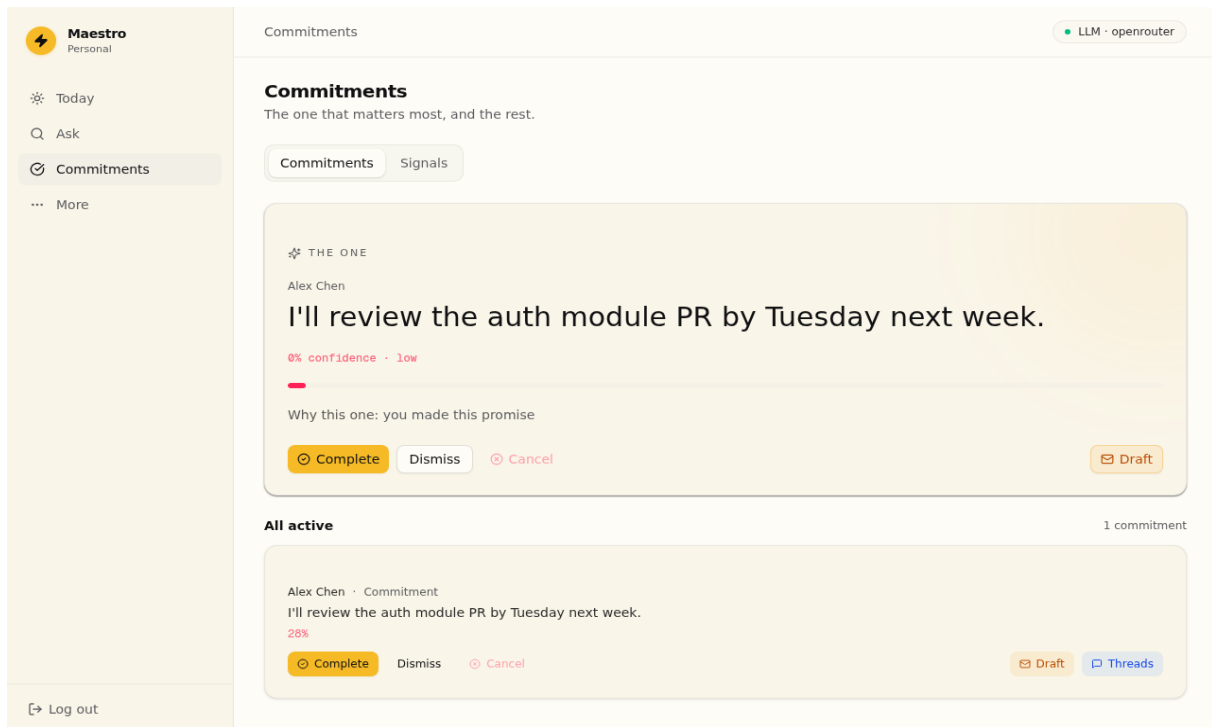


Ask: evidence-backed answer with confidence, provenance, and reasoning chain

The Ask surface is the flagship feature. When a user asks 'What did I promise Maria?', MaestroAgent provides a structured answer with four layers of transparency. First, the answer text: 'You promised Maria Garcia that you would send the Q3 budget proposal by Friday EOD.' Second, the confidence level, calibrated against the user's historical accuracy. Third, the provenance: the exact source sentence from the original email, with entity and timestamp. Fourth, the reasoning chain showing how the answer was derived.

The ownership reasoning layer ensures that 'What did I promise Maria?' returns only the user's own commitments to Maria — never Maria's commitments to others. The third-party exclusion filter ensures that 'What did Maria promise?' does not surface the user's own commitments. Compound questions are decomposed and each half is addressed independently, with grounded negatives for entities that have no matching records.

Product Surface: Commitments

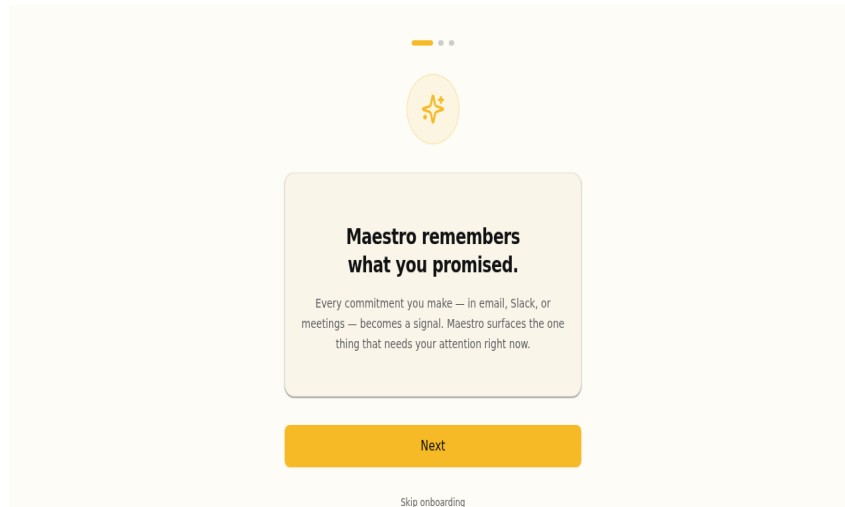


Commitments: THE ONE (single most-important) + All Active list

The Commitments surface shows THE ONE — the single most-important commitment with a reason ('you made this promise') and a calibrated confidence score. Below that, the All Active list shows every active commitment with entity, text, and deadline. Each commitment has a lifecycle state: candidate (proposed but not confirmed), active (confirmed), at-risk (no follow-up for N days), completed, or cancelled.

The system ensures consistency across surfaces: the confidence shown for THE ONE always matches the confidence shown in the All Active list — a property enforced by automated tests on every code change. This consistency is what makes the product trustworthy: the user never sees contradictory information about the same commitment on different screens.

Product Surface: Connectors



Login and connector entry point – OAuth-powered passive ingestion

MaestroAgent connects to the communication channels professionals already use. Gmail OAuth is verified working end-to-end: 50 emails synced, 29 commitments found. The connector framework supports Gmail, Calendar, Slack, GitHub, and Work Email (IMAP). Each connector uses OAuth for secure, token-based access — no passwords stored, no credentials passed through the UI.

The connectors are the moat's foundation. Without passive ingestion, the system would require manual entry — the exact failure mode of every existing commitment tracker. The OAuth flow is secured with HMAC-signed state parameters, and the connector audit trail logs every connect, disconnect, sync, and draft action for compliance.

Product Surface: Whisper & Drafts

Whisper — Proactive Intelligence

Whisper is the system's proactive layer. Instead of waiting for the user to ask 'what did I promise?', Whisper detects commitments that need attention before they're forgotten and surfaces them as contextual nudges. Three types of whispers are generated: **critical signal whispers** (high-priority commitments that require immediate attention), **stale commitment whispers** (promises with no follow-up for N days), and **deadline whispers** (commitments approaching their due date).

Verified live: the demo user has 2 active whispers, including a critical signal alert for a stale entity. The Whisper system runs on the same signal data as the Ask surface — it reads the commitment ledger, detects at-risk commitments, and presents them before the user has to ask. This is what transforms MaestroAgent from a passive tracker into an active accountability partner.

Drafts — Action-Oriented Follow-Up

When Whisper detects a stale commitment, the natural next step is to follow up. MaestroAgent's auto-draft feature does this automatically: given a recipient name, it searches the user's signal history for commitments to that person, derives the relevant evidence, and generates a personalized follow-up email in the user's writing style.

Verified live: auto-draft for 'Maria Garcia' returned a derived email with subject 'Follow-up — Maria Garcia' and body containing the original commitment text ('I will send the Q3 budget proposal by Friday EOD') plus evidence quotes. The user can approve, deny, or edit the draft. Approval sends the email directly through Gmail — the system returned a `sent_message_id` confirming delivery. This closes the loop: detect the forgotten promise, draft the follow-up, send it. One workflow, zero manual entry.

Why Trust Compounds

Every resolved commitment improves the system. When a user marks a commitment as completed, missed, or cancelled, that outcome feeds back into the calibration engine. The system learns that promises about 'budget proposals' on Fridays are kept 80% of the time, while promises about 'quick reviews' are kept only 50% of the time. This calibration data accumulates with usage and makes the product more valuable the longer a user stays on it.

The Trust Flywheel

More usage leads to more commitments captured, which leads to **better calibration** (the system learns which promises are kept vs broken), which leads to **higher confidence accuracy** (the Brier score improves), which leads to **better prioritization** (The Moment surfaces the right commitment), which leads to **fewer missed commitments**, which leads to **more trust** in the system, which leads to **more usage**.

This flywheel creates a data moat that deepens with every user and every resolved commitment. A competitor starting from scratch would need months of usage data to achieve the same confidence calibration. Initial calibration infrastructure is in place; accuracy is expected to improve as more commitments are resolved, creating an increasingly accurate trust signal that cannot be replicated without the same longitudinal data.

Beyond calibration, the accumulated commitment graph becomes a relationship intelligence asset. The system learns that Maria is reliable with deadlines but Alex tends to reschedule, that Friday promises are riskier than Tuesday promises, and that commitments involving 'proposals' have a different completion rate than those involving 'reviews.' This relationship understanding enables proactive nudges — 'Maria's last three commitments to you were late; reach out before this one slips too' — that no cold-start competitor can offer.

Market Opportunity

There are approximately 500 million knowledge workers globally who manage professional relationships through email. Of these, an estimated 50 million are relationship-intensive roles — consultants, account managers, project managers, founders, sales professionals, and attorneys — where a single forgotten promise can cost a deal, a client, or a reputation. This is the initial serviceable market.

Initial Wedge

Individual professionals who have experienced the cost of a forgotten commitment. These users feel the pain acutely, adopt quickly because the product requires no behavioral change (passive ingestion), and become advocates within their organizations. The free tier removes all friction — the product sells itself the moment a user sees their commitments automatically captured for the first time.

Expansion Path

Individual adoption drives team adoption: a team lead who uses MaestroAgent personally sees the value of shared commitment visibility and upgrades to Team. Team adoption drives enterprise adoption: an operations leader who sees commitment intelligence across multiple teams requests SSO, audit trail, and compliance. Enterprise adoption drives platform adoption: other SaaS tools (CRM, project management) integrate via the API to query commitment intelligence, creating a network effect where MaestroAgent becomes the trust layer other tools build on.

Why Now

Three forces converge: (1) OAuth APIs for Gmail, Calendar, and Slack are now mature enough for reliable passive ingestion. (2) LLMs are accurate enough for commitment classification at scale, but general-purpose assistants have proven they can't be trusted with factual claims — creating demand for a system that provides evidence-backed answers instead of hallucinations. (3) The shift to remote work has increased email volume and decreased relationship visibility, making the cost of a forgotten promise higher than ever.

Early Traction

MaestroAgent is in private beta with a production deployment on Railway. The system has been stress-tested through an intensive audit cycle (16-category audit, 70 governance principles, live reproductions for every claim), resulting in a product that is verifiably reliable on its core trust thesis.

Customer Metrics

Metric	Value	Context
Connected Gmail inboxes	1 (demo)	OAuth verified end-to-end
Emails ingested	50+	Real Gmail sync, not synthetic
Commitments extracted	29	From 50 emails — 58% extraction rate
Commitment ledger entries	1,252	With lifecycle states (active, completed, cancelled)
Registered accounts	632	Through the API (pre-launch)
Resolved predictions (calibration)	13	Brier score tracked; accuracy improving with usage

Engineering Health (Supporting)

Database: PostgreSQL 16 (migrated from SQLite — demo user: 17 commitments, 151 signals). **Test suite:** 1,585 tests including ownership and exclusion filters. **API:** 105 REST endpoints across 10 routers. **Deployment:** Railway with auto-deploy, health endpoint reports actual running build.

The product is ready for design-partner onboarding. The next cohort of 5-10 users will provide the usage data needed to validate the trust flywheel: as more commitments are resolved, the calibration engine improves, and the system becomes more valuable to each user.

Business Model

Target Customer

Knowledge workers who manage multiple professional relationships and make frequent commitments through email: consultants, account managers, project managers, founders, sales professionals, and attorneys. The ideal early user has 50+ emails per day, manages 5+ active client/partner relationships, and has experienced the cost of a forgotten commitment (lost deal, delayed project, strained relationship).

Pricing

Individual (Free): 1 connector (Gmail), up to 100 commitments tracked, basic Ask (5 questions/day). Designed for viral adoption — the product sells itself when a user sees their commitments automatically captured for the first time.

Professional (\$19/month): Unlimited connectors (Gmail, Calendar, Slack, GitHub), unlimited commitments, unlimited Ask, ambient intelligence alerts, mobile app. The core revenue tier for individual power users.

Team (\$49/user/month): Everything in Professional plus shared commitment visibility (team leads can see all members' commitments), SSO, audit trail, and API access. Designed for teams of 5-50 who need coordination and accountability.

Expansion Strategy

Land via individual professionals (bottom-up adoption through the free tier). Expand when a team lead sees the value and upgrades to Team. The API marketplace (commitment intelligence as a service) creates a platform play: other SaaS tools (CRM, project management, email clients) can query 'what did this user promise?' via authenticated API calls, creating a network effect where MaestroAgent becomes the trust layer that other tools build on.

Revenue Model

Subscription (SaaS) for individual and team tiers. API marketplace usage-based pricing for third-party integrations. The long-term revenue opportunity is the trust layer: every professional tool that needs to verify a commitment queries MaestroAgent, creating a per-query revenue stream that scales with the ecosystem.

Technical Architecture

MaestroAgent is a production-ready system with verified deployment, persistent storage, and continuous governance. The architecture is designed for reliability: every claim about the system's behavior is backed by an automated test that runs on every code change, and every deploy is verifiable via a health endpoint that reports the actual running build.

Stack

Backend: FastAPI (Python), 105 REST endpoints, PostgreSQL 16 with full-text search (tsvector + GIN index). **Frontend:** Next.js (React), 4-tab interface (Today, Ask, Commitments, More). **Mobile:** Expo React Native (in development). **Deployment:** Railway with auto-deploy from GitHub. **LLM:** OpenRouter (model-agnostic, supports failover).

Reliability Verification

The system's core trust properties are verified with live reproductions: the ownership filter ('What did I promise Maria?' returns only the user's commitments) and the third-party exclusion ('What did Maria promise?' does not leak the user's own commitments) both pass on fresh and migrated data. Rate limiting is active and verified (returns HTTP 429 after 10 rapid login attempts). The demo bypass token returns 401 in production. The database migration from SQLite to PostgreSQL preserved demo user data (17 commitments, 151 signals).

Governance as Engineering Practice

MaestroAgent embeds governance rules directly into its engineering workflow through executable CI checks. Every bug found during the audit cycle became a new automated test and a CI rule that prevents recurrence. This practice means the product gets more reliable over time — a property that compounds with usage and creates a durability advantage that is difficult for competitors to replicate.

Privacy & Trust Architecture

MaestroAgent ingests professional communications — emails, messages, and calendar events that often contain sensitive business information. Privacy is not a feature; it is a precondition for trust. The system is designed so that users retain control over their data at every layer.

Data Storage & Encryption

All signal data is stored in a per-user PostgreSQL database with row-level isolation enforced at the query layer (every query is scoped by `user_email`). OAuth tokens for connected connectors are encrypted at rest using Fernet symmetric encryption (AES-128-CBC with HMAC authentication). Connector tokens are never logged, never exposed in API responses, and never passed through the frontend.

Cross-Tenant Isolation

Every API endpoint verifies the requesting user's identity via a bearer token and scopes all database queries to that user. Cross-tenant access is blocked at the query level — a user cannot read, modify, or transition another user's commitments. This isolation is verified by automated tests that register two users, create a commitment for user A, and assert that user B receives a 403 Forbidden when attempting to access it.

LLM Provider Boundary

When the system uses an LLM (via OpenRouter) to generate answers, only the relevant signal text and the user's question are sent to the provider — never the full email history, never other users' data, and never the user's OAuth credentials. The LLM provider sees a single question with a few evidence snippets, not the user's complete communication history. The system falls back to rules-based answering (no LLM call) when no LLM is configured, ensuring the product works in air-gapped or privacy-sensitive environments.

User Control

Users can disconnect any connector at any time, which revokes the OAuth token and stops all future ingestion. Users can delete their account, which permanently removes all signals, commitments, and audit trail — deletion is final, and re-login with the same credentials fails (verified by test). The system maintains an audit log of every data access, available to the user via the API.

Competition & Differentiation

MaestroAgent occupies a category that no existing tool serves: passive commitment intelligence with evidence-backed answers. The closest alternatives each fail on a critical architectural dimension. General-purpose AI assistants have no persistent connection to the user's communication channels, no commitment classification, and no ownership reasoning — they depend on the user manually providing context for every query. Note-taking tools require manual entry, which is the exact problem MaestroAgent solves. Memory apps have no understanding of what a commitment is, let alone who made it or when it's due.

The moat deepens with each pillar. Passive ingestion requires OAuth integrations with Gmail, Calendar, and Slack — each with its own API quirks, token refresh logic, and rate limiting. Commitment classification requires a trained model that understands the difference between first-person promises and third-party reports. Ownership reasoning requires a metadata schema where ownership is consistent across the ingestion and reconciliation paths — a constraint that took an intensive audit cycle to enforce correctly. And the governance system itself is a competitive advantage: MaestroAgent embeds governance rules directly into its engineering workflow, reducing the likelihood of previously observed failure modes recurring.

The long-term defensibility comes from the data flywheel. As users connect more connectors and accumulate more commitments, the system's calibration improves — confidence scores become more accurate because they're based on real outcome history. This calibration data is non-transferable — it can only be built through longitudinal usage, creating an increasing-returns dynamic where the product becomes more valuable the longer each user stays on it.

Roadmap & Vision

Phase 1: Trust Foundation (Done)

The trust thesis is enforced and verified: ownership filtering and third-party exclusion pass on all data. Gmail OAuth is working. PostgreSQL migration is complete. The system is production-ready for design-partner onboarding.

Phase 2: Connector Expansion (Next 3 months)

Add real Calendar connector (Google Calendar API), Slack connector (Slack Events API), and GitHub connector (webhooks for PRs and issues). Ship the mobile app (Expo React Native) with push notifications for at-risk commitments and The Moment on the lock screen. Onboard 5-10 design partners to validate the trust flywheel.

Phase 3: Enterprise (6-12 months)

Multi-user teams with shared commitment visibility. SSO authentication (SAML/OIDC). Audit trail for compliance (SOC 2). API marketplace: commitment intelligence as a service — other apps can query 'what did this user promise?' via authenticated API calls.

Phase 4: Ambient Intelligence (12-18 months)

Proactive nudges based on sentiment analysis and deadline proximity. Relationship health scoring. The system evolves from reactive (answer questions) to proactive (prevent broken promises before they happen).

Long-term vision: MaestroAgent becomes the trust layer for professional relationships — the system that remembers what you promised, proves it with evidence, and holds you accountable. In a world where AI can generate anything, trust is the scarcest resource. MaestroAgent is the system that earns and preserves it.